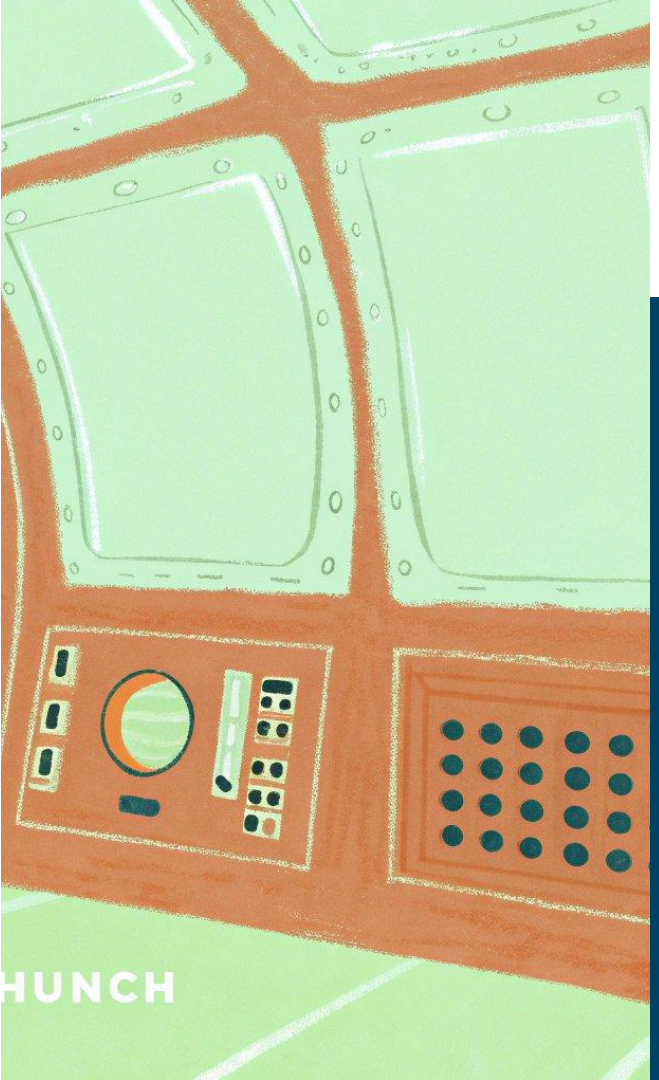


Lunar Smart Adaptive Habitat Design with AI Psychological Monitoring for Crew Adaptive Positive Mental Health

2025-26

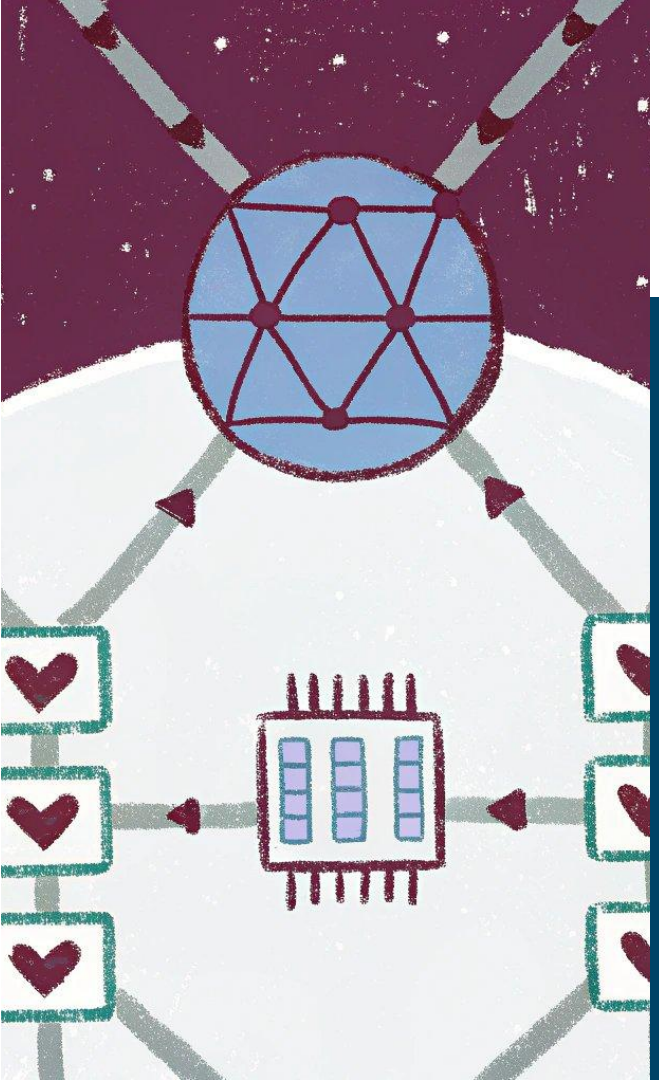
BIOMEDICAL
And / Or
Software Projects or Both



Lunar Smart Adaptive Habitat Design with AI Psychological Monitoring for Crew Adaptive Positive Mental Health

Goal:

Create a lunar habitat that uses **Psychological Research and** AI to monitor and support astronaut mental health, while integrating smart building systems for life support and power.

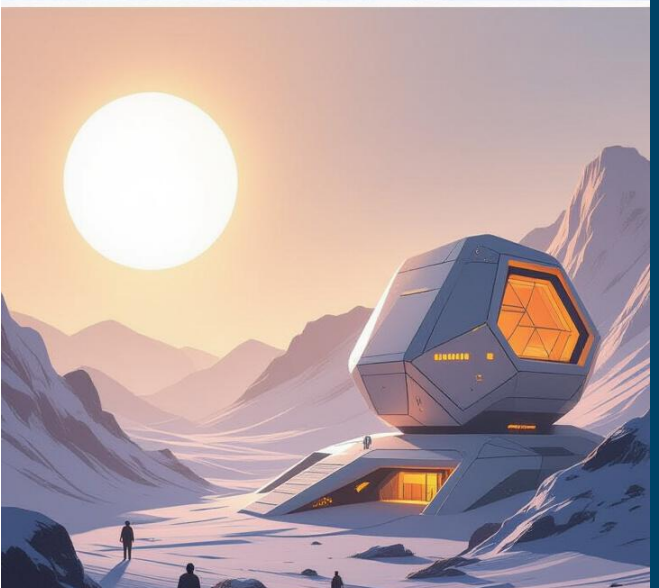


Project Overview

Mission:

Support astronaut **mental health** on the Moon NASA HUNCH empowers students to solve real NASA challenges. This project combines software biomedical (health, well-being) and (AI, monitoring) and approaches to design a next-generation lunar habitat.





The Problem

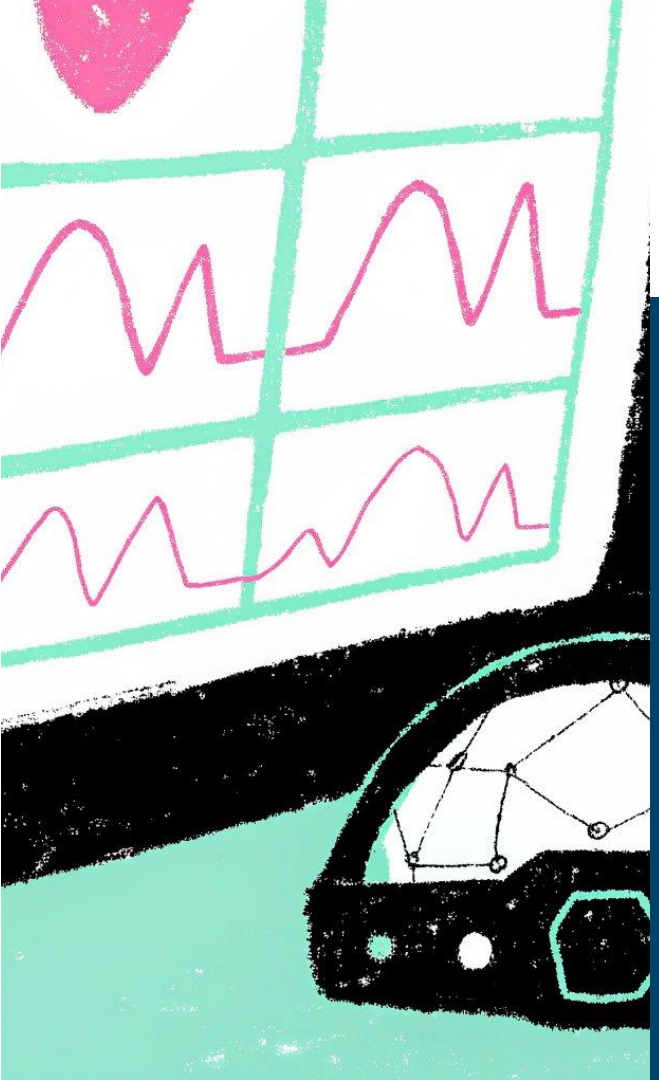
Long lunar missions = isolation, stress, and risk to mental health Current habitats: focus on physical safety, not psychological needs Need: Adaptive, supportive environments for crew well-being Astronauts face unique psychological challenges: isolation, confinement, and stress. Traditional habitats protect physically, but we must also address mental health for mission success.



The Challenge

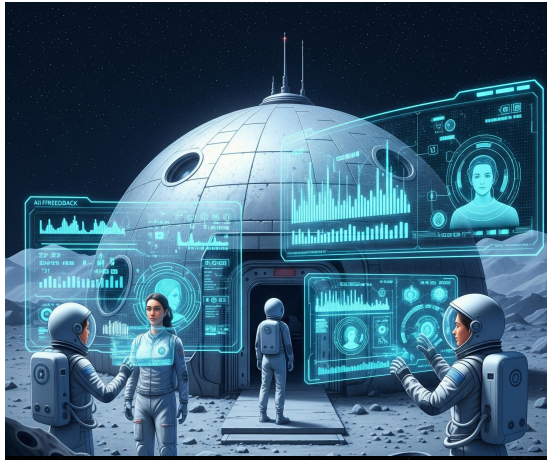
Design a lunar habitat that: Monitors crew psychological health with AI Adapts environment to individual needs Integrates smart building systems (ECLSS, power, environment) Is feasible for lunar deployment Our challenge: blend AI, biomedical monitoring, and smart building tech into a habitat that senses, predicts, and responds to each astronaut’s needs—while working reliably on the Moon.

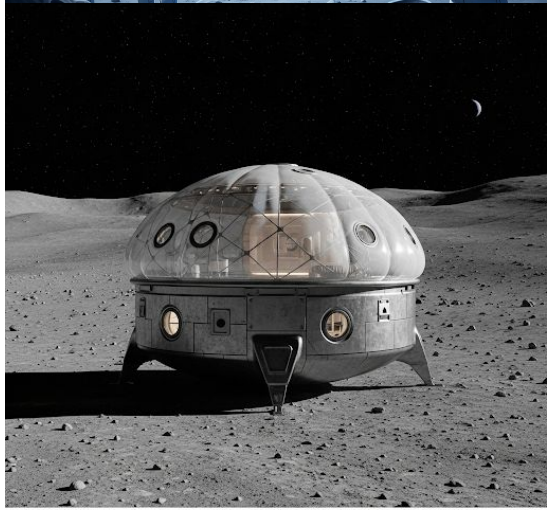
Provide an actual working demo of Personal Space and some Common Spaces for



Deliverables - Creative Problem Solving

- AI psychological monitoring dashboard
- Adaptive feedback/intervention system
- Biomedical Simulation: Simulated wearable data (heart rate, sleep, mood)
- Personalized countermeasures
- Scale Model 3D Printed habitat, Foam Board Habitat and electronic sensor systems with real time monitoring and reporting.
- Students will deliver a working prototype, a biomedical simulation, or a physical or virtual model.





So What would I do First?

- Research NASA Pill Habitat Design for the moon

Feature	Details (Based on NASA Artemis/Surface Habitat Concepts)
Crew Capacity	2 astronauts (standard), expandable to 4 (peak)
Uncrewed Duration	Up to 6–24 months of autonomous operation
Structure	Multi-level: rigid metallic base + inflatable upper deck
Internal Layout	Sleeping, food prep, hygiene, work, storage
Life Support	Advanced ECLSS, waste management, radiation shielding
Power	Solar arrays + nuclear fission backup
Location	Lunar south pole (e.g., Shackleton Crater)
Autonomy	Remote monitoring, automated maintenance, and mode switching

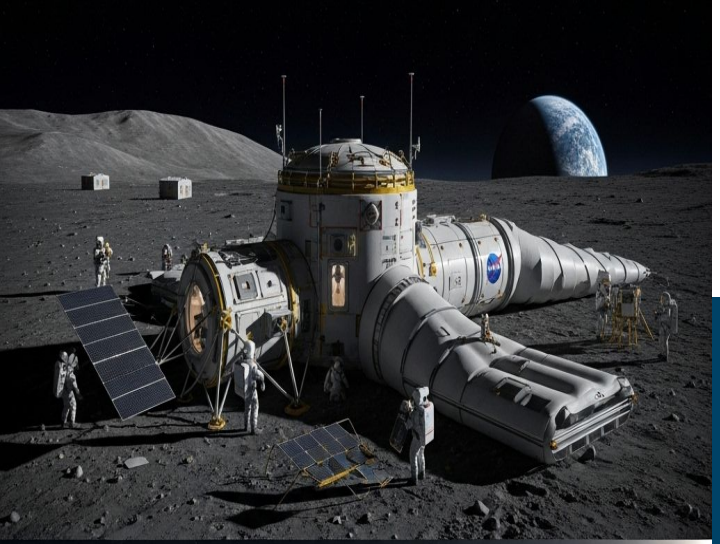
Research Continuous

1- Psychological issue Astronauts could have on Long Duration Mission

2- Research the ECLIS System for Human Habitats on the moon.

3- Research Smart Adaptive Habitat Design with AI Psychological Monitoring

https://docs.google.com/document/d/1QYRU6gYZP41-t6Jid_vsEwuNQygV9c91Uw7aOOprhUY/edit?usp=sharing





Make a Plan and ....

Sketch out a few designs for the outside

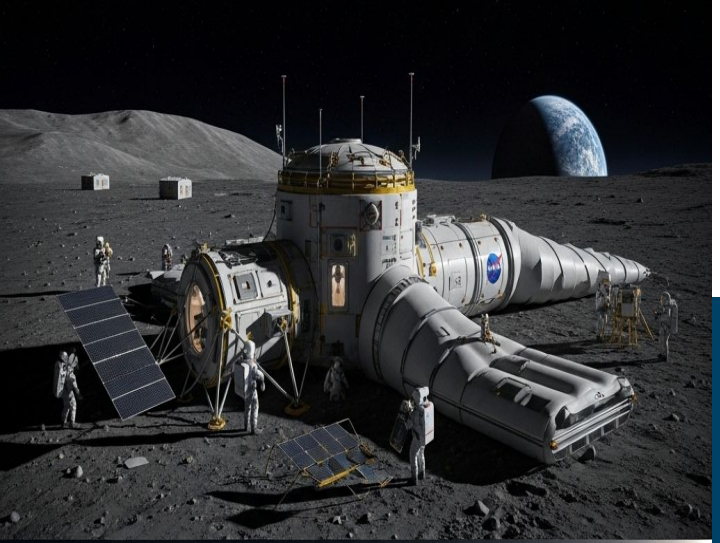
Build a Scale Model on a 2ft by 2ft piece of wood

- Foam Board
- Cardboard
- 3D Print

Make sure the roof or floors can be removed to look inside

Make a separate box outside to hold your electronics (Not Part of the Building.

Focus on the Layout





Make a Plan and ....

Sketch out a few designs for the outside

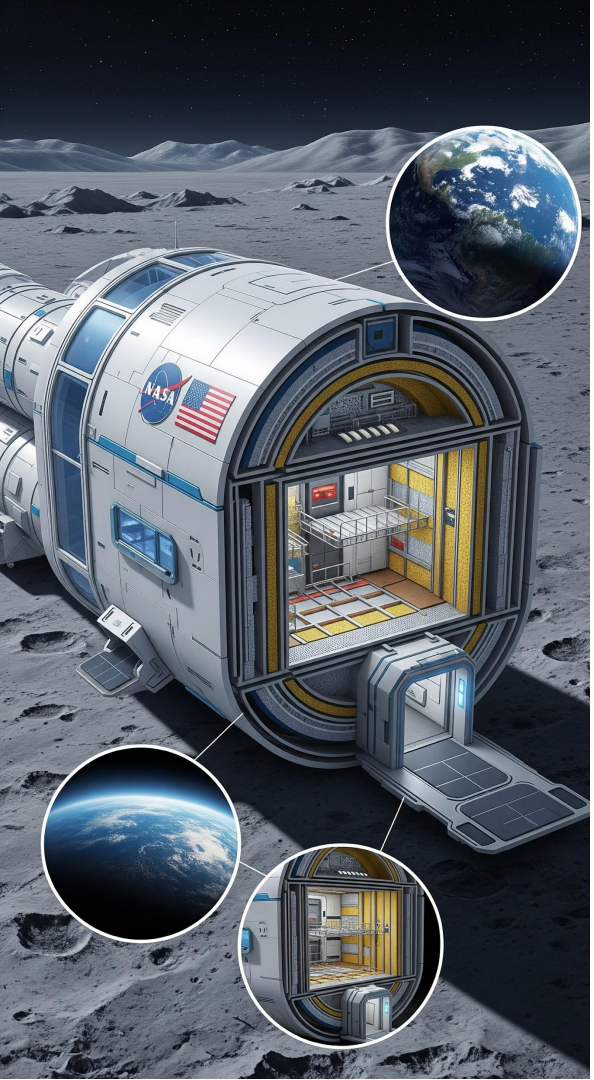
Build a Scale Model on a 2ft by 2ft piece of wood

- Foam Board
- Cardboard
- 3D Print
- Other

Make sure the roof or floors can be removed to look inside

Make a separate box outside to hold your electronics (Not Part of the Building.

Focus on the Layout

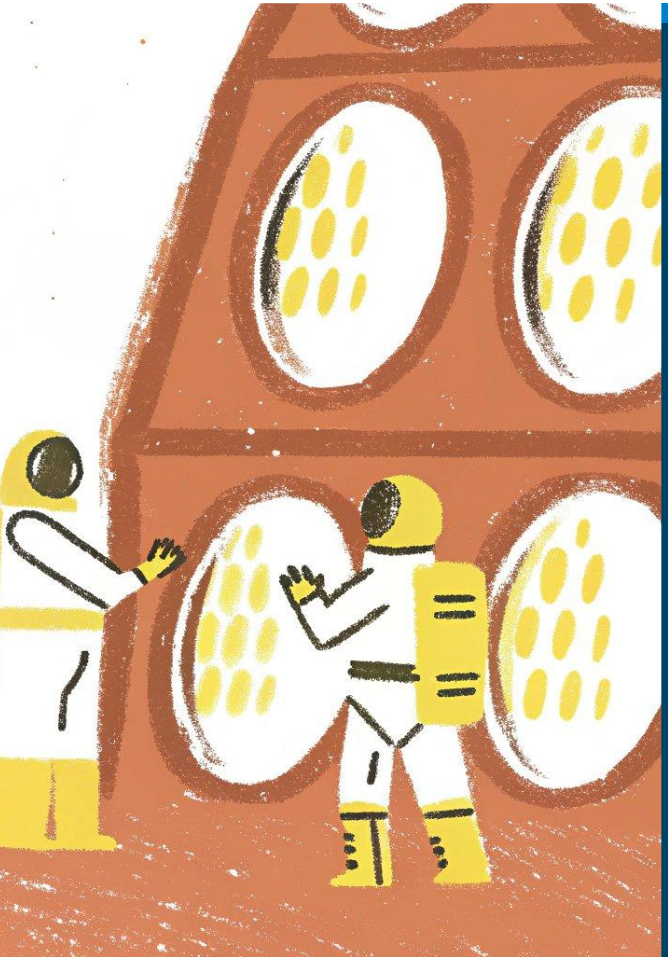


Make your Lunar building …

- Smart
- Adaptive Habitat Design with AI

- With Psychological Monitoring (Voice, Facial, Group Interaction, Lafter, Jokes, tears, isolation)

- Focus on one Room at a time



Key Considerations

Design considerations for a space habitat:

- **Dimensions**: ~20–120 m^3 per crew member
- **Structure**: Modular, expandable, radiation-shielded
- **AI Monitoring**: Real-time, privacy-respecting, adaptive
- **Smart Systems**: ECLSS, environmental, and power controls
- **Psychological Support**: VR/AR, adaptive lighting, social prompts

Must provide a Physical Design demonstrating Individual Space and Some Common Space with the application of destressors and feedback.

Design must balance comfort, safety, and efficiency. AI must be ethical and respect privacy. Smart systems should be robust and easy to use, supporting both physical and mental health.

More Info at:
https://docs.google.com/document/d/1QYRU6gYZP41-t6Jid_vsEwuNQygV9c91Uw7aOOprhUY/edit?usp=sharing

Constraints & Hints



Lunar Environment Constraints

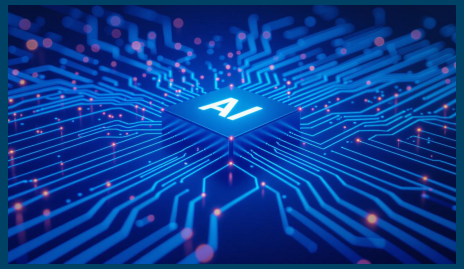
The **lunar environment** presents several constraints: **extreme temperatures**, **radiation**, pervasive **dust**, and **limited resupply** options from Earth. Designs must be resilient and resourceful to overcome these challenges, focusing on **autonomy** with minimal Earth intervention and efficient use of **lunar materials** and power. **Scalability** is essential for accommodating larger crews.



Design Hints

Leverage research from **ISS/Artemis habitats** and utilize **open-source tools** like **Unity** and **Arduino** to expedite development. Prioritize **user experience** to ensure designs are astronaut-friendly. The **Moon is harsh**, so designs must be **modular**, **autonomous**, and **efficient**. Focus on available technology to enhance the astronaut experience while ensuring resilience.







Example Project Flow

Research

Conduct thorough **research** on **psychological factors**, **lunar habitats**, and **smart systems** to lay a solid foundation for the project.

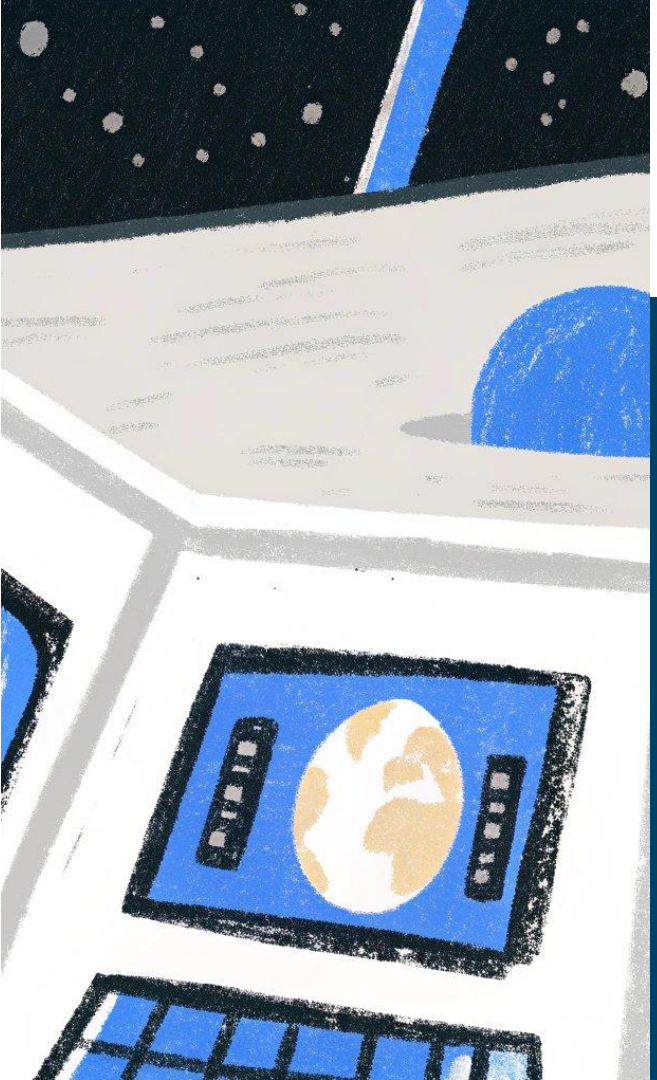
More Info at:
https://docs.google.com/document/d/1QYRU6gYZP41-t6Jid_vsEwuNQygV9c91Uw7aOOprhUY/edit?usp=sharing

Design

Create a detailed **design** by sketching the **layout**, defining **AI parameters**, and planning **integration** to guide the development process.

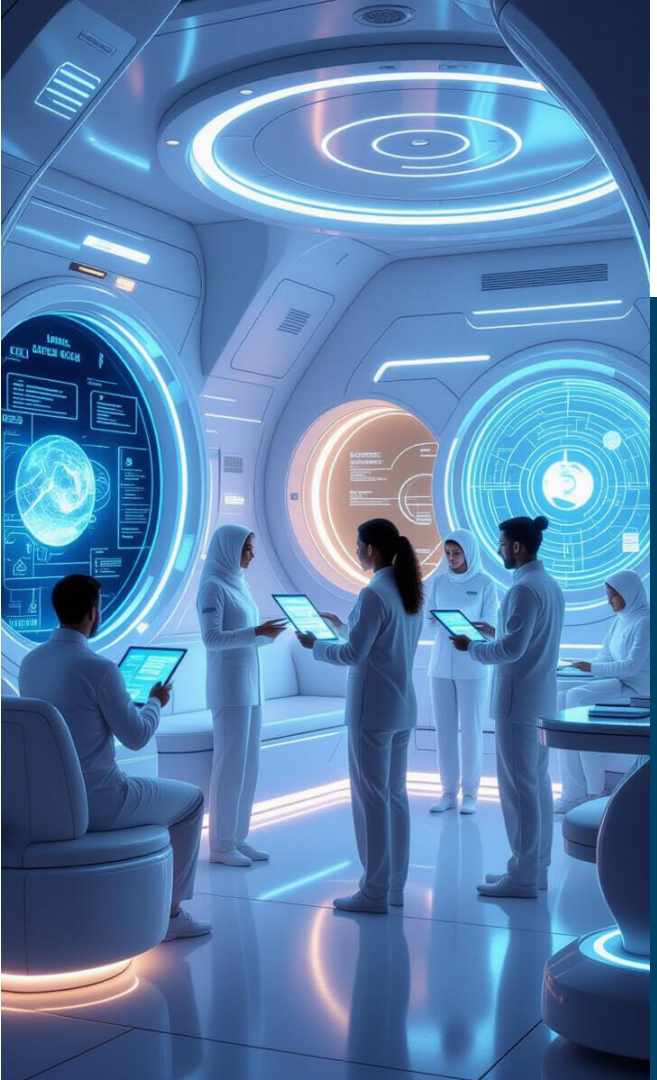
Prototype, Test, Present

Develop a **model/VR** prototype, **develop software**, **simulate scenarios**, **refine interventions**, and then **present** a **demo**, **walkthrough**, and **report**.



Conclusion

Students innovate for real NASA needs Combining, biomedical, software and engineering skills Supports future lunar explorers’ mental and physical health This project is a chance to make a real impact on the future of space exploration. By supporting astronaut well-being, you’re helping pave the way for humanity’s next giant leap.



Questions & Discussion

Ready to launch your ideas?

Let’s discuss your concepts and questions!!

More Info at:

https://docs.google.com/document/d/1QYRU6gYZP41-t6Jid_vsEwuNQygV9c91Uw7aOOprhUY/edit?usp=present

Other Details

First Floor

- **Airlock/Entry**
- **Storage**
- **Laboratory/Workspace**
- **Hygiene Area**

Second Floor

- **Crew Quarters (individual sleeping pods)**
- **Galley/Wardroom (dining and social area)**
- **Additional Storage**

The spatial organization separates noisy (work/lab) and quiet (sleeping) areas, with hygiene and galley areas placed for easy access.



Other Details - Room/Space - Recommended Dimensions (approximate)

Crew Quarters

2.1 m x 2.1 m x 2.4 m (7' x 7' x 8')

Galley/Wardroom

3.0 m x 3.0 m x 2.4 m (10' x 10' x 8')

Hygiene Area

2.0 m x 2.0 m x 2.4 m (6.5' x 6.5' x 8')

Laboratory/Workspace

4.0 m x 3.0 m x 2.4 m (13' x 10' x 8')

Storage

2.0 m x 2.0 m x 2.4 m (6.5' x 6.5' x 8')

Airlock

2.0 m x 2.0 m x 2.4 m (6.5' x 6.5' x 8')

Include: EVA Suit and Vehicle Port



Research

Key for this project is to start to Research in:

- **Psychological Support**: VR/AR, adaptive lighting, social prompts
- **AI Monitoring**: Real-time, privacy-respecting, adaptive, Psychological and Physical
- **Smart Systems**: ECLSS, environmental, and power controls
- **Structure**: Modular, expandable, radiation-shielded

